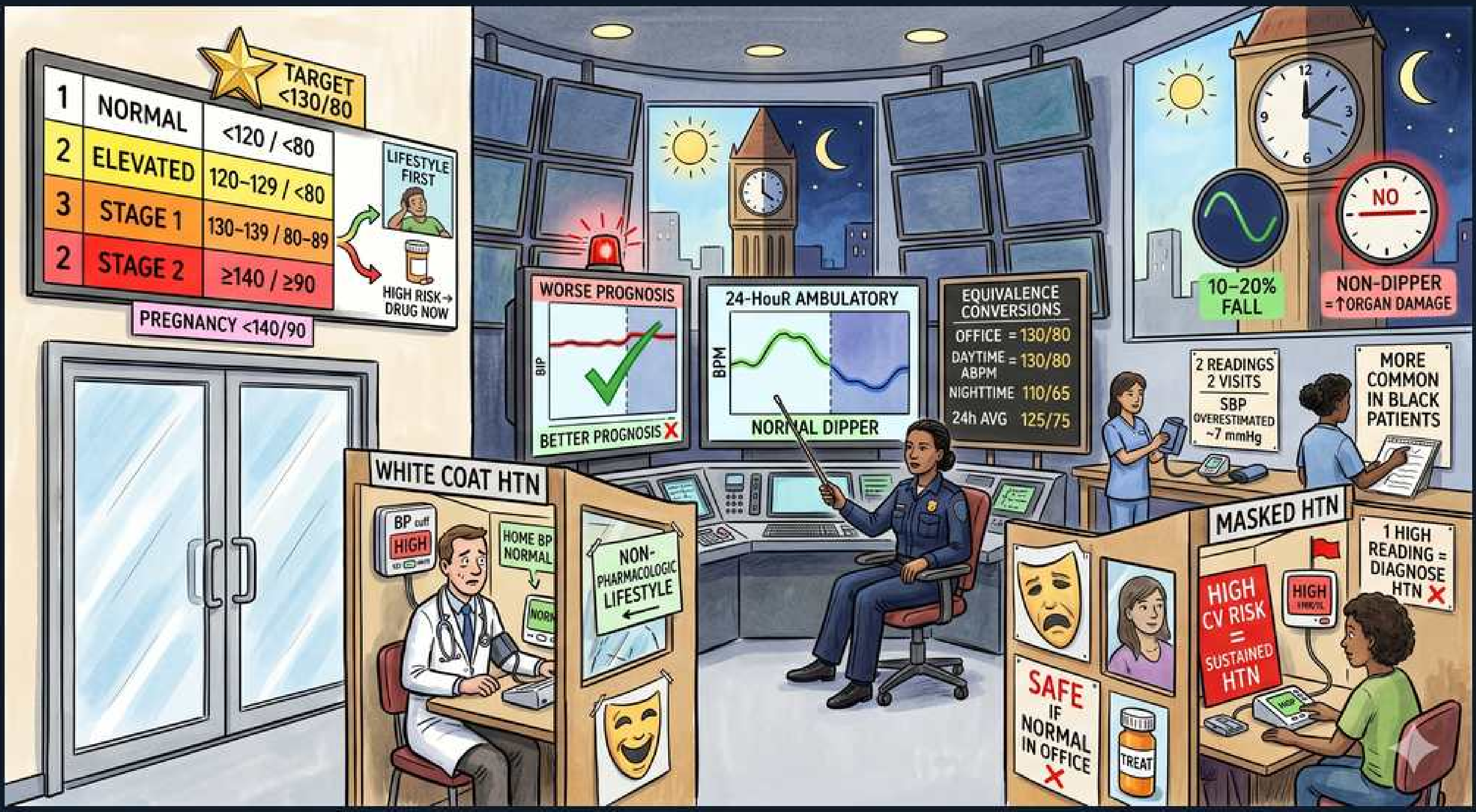
Hypertension — Classification, Targets & White-Coat/Masked HTN



1. BP staging chart

WHAT YOU SEE

Color-coded table: Normal <120/80, Elevated 120-129/<80, Stage 1 130-139/80-89, Stage 2 >=140/>=90.

WHAT IT ENCODES

ACC/AHA 2017 staging is the testable framework: Normal <120 AND <80; Elevated 120-129 AND <80; Stage 1 = 130-139 OR 80-89; Stage 2 = >=140 OR >=90. The OR is key — either the systolic OR the diastolic crossing the threshold defines the stage, so 134/76 is Stage 1 and 142/78 is Stage 2. Diagnosis requires the average of >=2 readings on >=2 separate occasions. Primary (essential) HTN accounts for ~90-95% of cases; suspect secondary causes when onset is <30 or >55, when it is resistant, or when there are red-flag features.

BOARD TRAP

A patient with BP 138/84 is labeled 'normal/borderline' — WRONG. By ACC/AHA criteria 130-139/80-89 is Stage 1 hypertension. The discriminator: do not use the legacy JNC7 cutoff of 140/90 to define disease; under current guidelines 140/90 is already Stage 2.

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2. Target <130/80

WHAT YOU SEE

Gold star at top reading TARGET <130/80.

WHAT IT ENCODES

The general BP goal for most adults is <130/80 mmHg, including patients with diabetes, CKD, established ASCVD, or age >=65 (if ambulatory and tolerating therapy). This target is supported by SPRINT, which showed intensive control (SBP <120 by automated office BP) reduced CV events and mortality versus a <140 target. Apply <130/80 broadly; tailor cautiously in frail elderly with orthostasis or limited life expectancy.

BOARD TRAP

Choosing a goal of <140/90 for a 68-year-old diabetic — WRONG. <140/90 is the obsolete JNC8 target; current ACC/AHA goal is <130/80 even for diabetics, CKD, and most elderly. The discriminator is that high CV risk demands the tighter target, not a looser one.

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3. Lifestyle first / high-risk drug now

WHAT YOU SEE

Pink tab: LIFESTYLE FIRST, then HIGH RISK -> DRUG NOW.

WHAT IT ENCODES

For Stage 1 HTN (130-139/80-89), management depends on ASCVD risk: if 10-year ASCVD risk <10% AND no comorbidity, use lifestyle modification alone with reassessment in 3-6 months. Start drug therapy at Stage 1 if 10-year ASCVD >=10%, OR diabetes, CKD, or known ASCVD/HF. ALL Stage 2 patients (>=140/90) get drug therapy plus lifestyle, and most start with two agents. Lifestyle = DASH diet, sodium <1500-2300 mg/day, weight loss (~1 mmHg per kg), exercise, alcohol limit, K-rich diet.

BOARD TRAP

Starting an antihypertensive for every Stage 1 patient — WRONG. A low-risk Stage 1 patient (ASCVD <10%, no comorbidity) is managed with lifestyle alone first. The discriminator: you must risk-stratify with the ASCVD calculator before reaching for a drug at Stage 1; the rule differs from Stage 2, where everyone gets pharmacotherapy.

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4. Pregnancy <140/90

WHAT YOU SEE

Purple banner: PREGNANCY <140/90.

WHAT IT ENCODES

In pregnancy the treatment threshold and target are higher (~140/90, often treating at >=140/90 to a goal around 130-150/80-100) to preserve uteroplacental perfusion. Safe agents = labetalol, nifedipine, and methyldopa (mnemonic 'Hypertensive Moms Love Nifedipine'). ACE inhibitors, ARBs, and direct renin inhibitors are ABSOLUTELY contraindicated (fetal renal dysgenesis, oligohydramnios, death). Severe-range BP >=160/110 in pregnancy is an emergency treated with IV labetalol, IV hydralazine, or oral nifedipine.

BOARD TRAP

Continuing lisinopril in a newly pregnant patient or aiming for <130/80 — WRONG on both counts. ACEi/ARBs are teratogenic and must be stopped, and the aggressive <130/80 target does not apply in pregnancy. The discriminator: switch to labetalol/nifedipine/methyldopa and accept a higher target to protect placental flow.

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5. Worse prognosis monitor

WHAT YOU SEE

Screen with rising red line, WORSE PROGNOSIS, BETTER PROGNOSIS marked.

WHAT IT ENCODES

Higher absolute BP and greater BP variability both independently predict worse cardiovascular outcomes — stroke, MI, HF, CKD progression, and retinopathy. Risk rises continuously from 115/75 upward, roughly doubling for every 20/10 mmHg increment. This is why even 'Elevated' BP (120-129) warrants lifestyle change: there is no truly safe ceiling, only escalating risk.

BOARD TRAP

Assuming a single low reading on one visit rules out cardiovascular risk — WRONG. Variability and the averaged trend, not one snapshot, drive prognosis. The discriminator: out-of-office averaging is needed to characterize true burden.

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6. 24-hour ambulatory monitor

WHAT YOU SEE

Monitor labeled 24-HOUR AMBULATORY with a normal dipping curve.

WHAT IT ENCODES

24-hour ambulatory BP monitoring (ABPM) is the GOLD STANDARD for diagnosis and best correlates with end-organ damage and outcomes. It is essential to confirm white-coat HTN (high office, normal out-of-office) and to detect masked HTN (normal office, high out-of-office). Home BP monitoring (HBPM) is an acceptable alternative when ABPM is unavailable. Diagnostic 24h-average threshold is >=125/75.

BOARD TRAP

Diagnosing and treating HTN off a single elevated clinic reading — WRONG. The discriminator: confirm with out-of-office monitoring (ABPM preferred, HBPM acceptable) before committing a patient to lifelong therapy, because office readings can reflect white-coat effect.

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7. Equivalence conversions

WHAT YOU SEE

Chart: Office 130/80, daytime ABPM 135/85, nighttime 110/65, 24h avg 125/75.

WHAT IT ENCODES

Diagnostic thresholds are setting-specific because out-of-office values run lower: office >=130/80 corresponds to daytime/awake ABPM or home >=135/85, nighttime/asleep >=110/65, and 24-hour average >=125/75. Memorize the 135/85 home cutoff — it is the most commonly tested non-office number.

BOARD TRAP

Applying the 130/80 office cutoff to a home reading of 132/82 and calling it hypertension — WRONG. The home/daytime-ABPM threshold is 135/85, so 132/82 at home is normal. The discriminator: match the threshold to the measurement setting.

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8. 10-20% nocturnal fall

WHAT YOU SEE

Clock with sun/moon and 10-20% FALL.

WHAT IT ENCODES

Normal circadian physiology produces a 10-20% nocturnal dip in BP during sleep ('normal dipper'). This nighttime fall reflects intact autonomic regulation. ABPM is the only way to capture the dipping pattern, since clinic readings cannot assess sleep BP.

BOARD TRAP

Interpreting a normal daytime clinic BP as reassuring while ignoring nighttime values — WRONG. The discriminator: the dipping pattern (only seen on ABPM) carries independent prognostic weight beyond daytime numbers.

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9. Non-dipper = organ damage

WHAT YOU SEE

Red NO clock: NON-DIPPER = ORGAN DAMAGE.

WHAT IT ENCODES

A 'non-dipper' (nocturnal fall <10%) — or a 'reverse dipper' whose BP rises at night — has more end-organ damage (LVH, microalbuminuria, CKD) and higher CV/stroke risk than a normal dipper. Non-dipping is associated with secondary HTN, OSA, CKD, diabetes, and autonomic dysfunction; identifying it can prompt evaluation for these and bedtime dosing strategies.

BOARD TRAP

Reassuring a patient whose clinic BP is at goal but who is a non-dipper on ABPM — WRONG. The discriminator: non-dipping signals occult risk and possible secondary cause (e.g., OSA), warranting further workup rather than reassurance.

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10. More common in Black patients

WHAT YOU SEE

Note: MORE COMMON IN BLACK PATIENTS.

WHAT IT ENCODES

Non-dipping, salt-sensitive, and low-renin hypertension are more prevalent in Black patients, who also have earlier onset, higher prevalence, and greater rates of HTN-related complications (stroke, HF, CKD/ESRD). This drives the drug-selection rule: in Black adults without HF or CKD, start with a CCB or thiazide (or both) rather than ACEi/ARB monotherapy.

BOARD TRAP

Choosing ACEi monotherapy as first-line for a Black patient with uncomplicated HTN — WRONG. The discriminator: ACEi/ARBs are less effective as initial monotherapy in this group (lower renin), so a CCB or thiazide is preferred unless there is a compelling indication such as CKD with albuminuria or HFrEF.

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11. White coat HTN

WHAT YOU SEE

Doctor's office, HOME BP NORMAL, NON-PHARMACOLOGIC LIFESTYLE, happy/sad masks.

WHAT IT ENCODES

White-coat HTN = elevated office BP (>=130/80) but normal out-of-office BP (<135/85 home/daytime) in an untreated patient, driven by the clinical setting. It carries near-normal but slightly elevated long-term CV risk, so management is lifestyle modification plus periodic monitoring with annual ABPM/HBPM to detect progression to sustained HTN — generally NOT drug therapy.

BOARD TRAP

Starting an antihypertensive for a patient with office 142/88 but home averages of 124/78 — WRONG. This is white-coat HTN; treating it risks symptomatic hypotension. The discriminator: confirm with ABPM/HBPM and manage with lifestyle and surveillance, not medication.

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12. 2 readings, 2 visits rule

WHAT YOU SEE

Sign: 2 READINGS, 2 VISITS; SBP overestimated ~7 mmHg.

WHAT IT ENCODES

Diagnosis requires the average of >=2 readings obtained on >=2 separate occasions, using proper technique (seated 5 min rest, back/arm supported, correct cuff size, no caffeine/smoking/exercise for 30 min). Improper technique systematically overestimates SBP by up to ~5-10 mmHg; automated office BP and out-of-office confirmation reduce this error.

BOARD TRAP

Labeling a patient hypertensive from one rushed reading taken right after they walked in — WRONG. The discriminator: technique errors and white-coat effect inflate a single reading; require averaged, properly measured, repeated readings before diagnosis.

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13. Masked HTN

WHAT YOU SEE

Right cubicle: MASKED HTN, 1 HIGH READING -> DIAGNOSE, HIGH CV RISK, SUSTAINED HTN.

WHAT IT ENCODES

Masked HTN = NORMAL office BP but ELEVATED out-of-office BP (>=135/85 home/daytime). It is dangerous because it is easily missed yet carries CV risk equivalent to sustained HTN. Suspect it with target-organ damage (LVH, CKD, retinopathy) despite normal clinic readings, or in high-risk patients; confirm with ABPM/HBPM and treat as true hypertension with both lifestyle and drugs.

BOARD TRAP

Telling a patient with LVH on echo but normal clinic BP that their hypertension is 'controlled' or absent — WRONG. Normal office BP does NOT exclude HTN. The discriminator: end-organ damage with normal office readings should trigger ABPM to unmask masked HTN, which then requires treatment.